# Multi-Team Scaling Scenarios

- **Scenario 1 - 2 Teams (24 employees in Total)**
    - Team A (12 primary + 2 cross = 14 total)
    - Team B (12 primary + 2 cross = 14 total)
    - Each team has 2 cross-members borrowed from the next team.

- **Scenario 2 - 4 Teams (48 employees in Total)**
    - Team A (12 primary + 2 cross = 14 total)
    - Team B (12 primary + 2 cross = 14 total)
    - Team C (12 primary + 2 cross = 14 total)
    - Team D (12 primary + 2 cross = 14 total)
    - Each team has 2 cross-members borrowed from the next team.

- **Scenario 3 - 8 Teams (96 employees in Total)**
    - Team A (12 primary + 2 cross = 14 total)
    - Team B (12 primary + 2 cross = 14 total)
    - Team C (12 primary + 2 cross = 14 total)
    - Team D (12 primary + 2 cross = 14 total)
    - Team E (12 primary + 2 cross = 14 total)
    - Team F (12 primary + 2 cross = 14 total)
    - Team G (12 primary + 2 cross = 14 total)
    - Team H (12 primary + 2 cross = 14 total)
    - Each team has ~2 cross-members borrowed from the next team.

- **Scenario 4 - 16 Teams (192 employees in Total)**
    - Team A (12 primary + 2 cross = 14 total)
    - Team B (12 primary + 2 cross = 14 total)
    - Team C (12 primary + 2 cross = 14 total)
    - Team D (12 primary + 2 cross = 14 total)
    - Team E (12 primary + 2 cross = 14 total)
    - Team F (12 primary + 2 cross = 14 total)
    - Team G (12 primary + 2 cross = 14 total)
    - Team H (12 primary + 2 cross = 14 total)
    - .....
    - Team P (12 primary + 2 cross = 14 total)
    - Each team has ~2 cross-members borrowed from the next team (20% cross-membership).

# 4 different scenarios for vacation generation:

- **Scenario 1 — Sequential Rotation (Base method)**
    - Employees take 30 consecutive vacation days, distributed sequentially throughout the year.
    - For example the first employee starts a 30 day block on January 1st and the following employee begins after the previous one ends (In this case on January 31st). If one of the employees have their vacation at the end of the year it will continue at the beginning of the year.
    - This guarantees that vacations days are evenly distributed

- **Scenario 2 — Random Two-Block Distribution**
    - Employees are assigned vacations on two random blocks of 15 days across the calendar year
    - Guaranteed that these two blocks won't overlap with one another (to make sure the total of vacation days is always 30)

- **Scenario 3 — Weekend Focused Short Breaks**
    - Considering employees can work every year during any day, this scenario was created to mimic a more real human vacation scenario
    - Employees take multiple 4–6-day breaks, distributed throughout the year and biased toward weekends
    - Each employee's set of breaks totals **30 days**, spread over ~6–8 mini-vacations spaced about 3 to 5 weeks apart.

- **Scenario 4 — Summer-Focused Staggered Vacations**
    - All employees' vacations are restricted to the **summer period (June 1 – August 31)**.
    - Each employee takes **one continuous 30-day vacation**, with start dates evenly distributed within this 3-month window.
    - Used to represent seasonal vacation spikes

# Minimum Templates Generation

To simulate the operational staffing requirements corresponding to each multi-team scenario, a **Minimums Template Generator** was developed.

## Scaling Methodology

For each target scenario (4, 8, and 16 teams),
The script performs proportional scaling and controlled variation as follows:

1. **Per-Team Scaling Factor**
   The number of required employees per team is scaled by the ratio of target to base headcount:

$$Scale\ Factor\ = \frac{Employees\ per\ Team\ (Target)\ (excluding\ shared)}{Employees\ per\ Team\ (Base)}$$

   For example, if each team in the base scenario has 10 employees and the target scenario has 7 per team, all the daily minimums are multiplied by **0.7**. After multiplying they are then transformed into integers (using banker's rounding [e.g 2.5→2, 3.5→4])

2. **Controlled Random Variation (±1)**

   To avoid the templates to be exactly the same per team, a small random ±1 perturbation was applied to a small fraction of days (≈4–5%). These perturbations are balanced meaning that the number of +1 and -1 changes are equal to guarantee that the scaling remains constant.

3. **Ideal Staffing**

$$Ideal(d, t, e)\ =\ Minimum(d, t, e)\ \ +\ 1$$

   To guarantee that if one of the members of a team has to miss a work day due to an unexpected situation the minimums are still covered.

**Total Complete Scenarios** -> **4** Multi-Team Scaling * Scenarios **4** Vacation Scenarios =

## 16 Complete Scenarios

**Considering we are testing ILP and CSP, this makes up for 32 algorithmic results (16*2)**

# Results

## ILP (Maximum time - 90 minutes)

Weights used - 100 minimums, 1 ideals

| | Time Taken | Minimuns not accomplished | Ideals not accomplished | % of minimums not filled | % of ideals not filled |
|---|---|---|---|---|---|
| 2Teams_Vacation1 | 1 min 14 sec | 103 | 488 | 2.82% | 8.36% |
| 2Teams_Vacation2 | 1 min 14 sec | 103 | 488 | 2.82% | 8.36% |
| 2Teams_Vacation3 | 1 min 16 sec | 103 | 488 | 2.82% | 8.36% |
| 2Teams_Vacation4 | 1 min 55 sec | 103 | 674 | 2.82% | 11.54% |
| 4Teams_Vacation1 | 9 min 06 sec | 215 | 976 | 2.94% | 8.35% |
| 4Teams_Vacation2 | 9 min 27 sec | 215 | 976 | 2.94% | 8.35% |
| 4Teams_Vacation3 | 8 min 24 sec | 215 | 976 | 2.94% | 8.35% |
| 4Teams_Vacation4 | 5 min 08 sec | 216 | 1378 | 2.95% | 11.79% |
| 8Teams_Vacation1 | 45 min 00 sec | 412 | 1952 | 2.82% | 8.35% |
| 8Teams_Vacation2 | 56 min 01 sec | 412 | 1952 | 2.82% | 8.35% |
| 8Teams_Vacation3 | 48 min 45 sec | 412 | 1952 | 2.82% | 8.35% |
| 8Teams_Vacation4 | 36 min 07 sec | 492 | 2783 | 3.36% | 11.91% |
| 16Teams_Vacation1 | | | | | |
| 16Teams_Vacation2 | | | | | |
| 16Teams_Vacation3 | | | | | |
| 16Teams_Vacation4 | | | | | |

# CSP (Maximum time - 90 minutes)

Weights used - 100 minimums, 1 ideals

| | Time Taken | Minimuns not accomplished | Ideals not accomplished | % of minimums not filled | % of ideals not filled |
|---|---|---|---|---|---|
| 2Teams_Vacation1 | 4 min 09 sec | 103 | 488 | 2.82% | 8.36% |
| 2Teams_Vacation2 | 5 min 44 sec | 103 | 491 | 2.82% | 8.41% |
| 2Teams_Vacation3 | 2 min 43 sec | 103 | 488 | 2.82% | 8.36% |
| 2Teams_Vacation4 | 13 min 13 sec | 103 | 681 | 2.82% | 11.66% |
| 4Teams_Vacation1 | 90 min 03 sec | 215 | 976 | 2.94% | 8.36% |
| 4Teams_Vacation2 | 90 min 03 sec | 215 | 976 | 2.94% | 8.36% |
| 4Teams_Vacation3 | 78 min 04 sec | 215 | 976 | 2.94% | 8.36% |
| 4Teams_Vacation4 | 90 min 04 sec | 216 | 1383 | 2.95% | 11.84% |
| 8Teams_Vacation1 | 90 min 13 sec | 412 | 1952 | 2.82% | 8.36% |
| 8Teams_Vacation2 | 90 min 13 sec | 412 | 1952 | 2.82% | 8.36% |
| 8Teams_Vacation3 | 79 min 08 sec | 412 | 1952 | 2.82% | 8.36% |
| 8Teams_Vacation4 | 90 min 13 sec | 492 | 2784 | 3.36% | 11.91% |
| 16Teams_Vacation1 | | | | | |
| 16Teams_Vacation2 | | | | | |
| 16Teams_Vacation3 | | | | | |
| 16Teams_Vacation4 | | | | | |